

BUDGET IMPACT ANALYSIS TOOL

User Guide

For spreadsheet_BIA_v4.0.xlsm

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2026. Health Technology Assessment Unit (UATS) - Hospital Alemão Oswaldo Cruz (HAOC)

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Contents and purpose of this guide

This User Guide provides detailed instructions for completing and interpreting the Excel-based Budget Impact Analysis (BIA) Tool.

It follows the workflow implemented in **spreadsheet_BIA_v4.0.xlsm** and is intended to support users in selecting the appropriate input pathway, documenting assumptions and data sources, and avoiding inconsistent combinations of inputs.

The tool is designed for five-year budget impact analyses comparing a reference scenario with up to three alternative scenarios. It accommodates up to four proposed technologies and up to ten comparators. It can be used for medicines, procedures, tests, or other health technologies, provided that the population, market-share, cost, and treatment-duration assumptions can be represented by the available input structures.

Important: The spreadsheet is a calculation framework. It does not determine which epidemiological, utilization, market-share, clinical, or cost inputs are appropriate for a specific decision problem. Users remain responsible for selecting, documenting, and justifying all input data and assumptions.

Recommended workflow

Stage	Workbook tab	Main task
1	Definitions	Define the analysis, target population, proposed technologies, comparators, and scenarios.
2	Mkt share	Set the first budget year and annual market shares for the reference and alternative scenarios.
3	Population	Estimate or directly enter the eligible population for each of the five years.
4	Costs	Enter technology-specific and general costs, or use the support cost table when the structured approach is not feasible.
5	Time on treatment	Specify label-based treatment duration, discontinuation/progression, and mortality/overall survival where relevant.
6	Results	Review the final population and budget results for each scenario.
7	SensitivityAn / Parameters / PSA	Explore deterministic and probabilistic uncertainty.

Workbook architecture

The workbook contains user-facing tabs and calculation/support tabs. The main data-entry workflow is performed in Definitions, Mkt share, Population, Costs, and Time on treatment. Results and uncertainty-analysis tabs summarize outputs. Data, Calculations, Calculations2, Calculations3, Parameters, and DSA support the underlying calculations and should not be modified unless the user understands the model architecture and intends to modify the model itself.

Budget Impact Calculator - Blank Template

< Introduction > Next: Definitions

General Information

This document was developed by the Unit of Health Technology Assessment (UATS) of the Hospital Alemão Oswaldo Cruz, through a Proadi-SUS partnership.

This spreadsheet is intended solely for educational and non-commercial purposes. Reproduction, distribution, or use for commercial gain without prior written permission is strictly prohibited.

Instructions for Use

Build the model by defining the number of scenarios, alternatives, and time horizon in the following tabs.
For further guidance, see the instruction manual.

 Calculated or automatically populated fields. **They should not be changed.**

 Customizable fields. Must be completed as instructed.

 Instructions.

  Buttons for navigating between tabs.

Rationale for using the spreadsheet

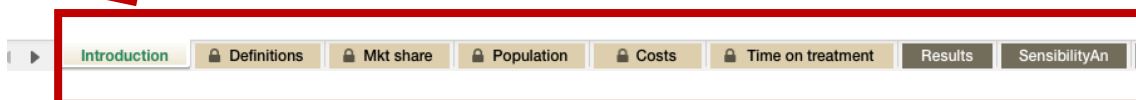
1. Save a copy of the spreadsheet to start a new analysis;
2. Under 'Definitions', describe the components of the research question: population (clinical condition), intervention(s), and comparators (alternatives);
3. and 4. Under 'Market Share' enter the year the analysis began and the expected market share percentages for each scenario (current and proposed scenarios);
5. Under 'Population', enter the method for calculating the eligible population and the parameters required for the estimates;
6. Under 'Costs', enter the resources and costs involved according to the type of technology;
7. Under 'Time of Use', indicate which strategy will be used to report the time of use of the technologies over the years;
- 8 and 9. Evaluate the results and sensitivity analyses in the 'Results' and 'SensitivityAn' tabs.

Date of last update of the template spreadsheet: September, 2026

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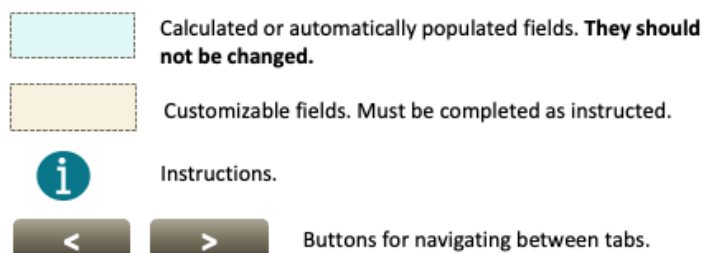
Validated by: Cecilia Menezes; Francino Azevedo; Juliana Viscondi; Mariana Fachi; Tassiane de Paula; Vinicius Ferreira



Tab	Role	Routine user action
Introduction	Workbook opening/navigation page.	Read instructions and use navigation controls where available.
Definitions	Analysis metadata and decision problem.	Complete required descriptive inputs and scenario names.
Mkt share	Time horizon and technology diffusion.	Enter first year and annual market shares.
Population	Eligible-population estimation.	Select one population-estimation pathway and enter supporting data.
Costs	Per-patient cost estimation.	Enter resource use, unit costs, utilization proportions, and references.
Time on treatment	Exposure and attrition over time.	Enter duration, discontinuation/progression, and mortality inputs as applicable.
Results	Base-case outputs.	Review population and budget-impact results; select dynamic-cohort option where appropriate.
SensitivityAn	Deterministic sensitivity analysis output.	Review DSA results/graphs after uncertainty parameters are defined.
PSA	Probabilistic sensitivity analysis output.	Review simulation results after PSA is run.
Data	Supporting time-series calculations and entry of non-annual data.	Use only when the selected discontinuation/progression/mortality option requires other data periodicities.
Parameters	Uncertainty parameterization.	Define variation/bounds, distributions, references, and DSA inclusion where needed.
Calculations / Calculations2 /	Internal calculations.	No routine data entry.

General data-entry conventions

The workbook uses protected worksheets to reduce the risk of accidental formula changes. In the principal input tabs, pale-yellow cells are used for user-entered values or selections, whereas other cells may contain formulas, labels, checks, or automatically populated information. Users should enter information only in intended input cells and **should not delete labels, formulas, or pre-populated content.**



General instructions

1. Save a new copy of the workbook before starting a new analysis.
2. Complete the tabs in the recommended order because technology names, scenario names, years, and other selections are carried forward to subsequent tabs.
3. Use the reference fields provided in the Population and Costs sections to document the source of each parameter whenever possible.
4. When an input is assumption-based rather than empirically estimated, identify it explicitly as an assumption in the analysis documentation.
5. Do not combine alternative population-estimation approaches unless the workbook specifically instructs you to do so.
6. Check that market shares sum to 100% in each year and scenario.
7. Do not enter percentages as counts or counts as percentages; follow the units indicated in each section.
8. Do not overwrite automatically calculated totals or summary tables.

1. Definitions: setting up the analysis

The Definitions tab establishes the decision problem and the labels used throughout the workbook. Complete this tab before entering quantitative data.

1.1. Analysis identification

Budget Impact Calculator - Blank Template

< Definitions > Next: Market share and time horizon

1. Starting a new budget impact analysis
When starting a new analysis, save a copy of this document and then complete the following fields.

Title of the Analysis	Budget Impact Calculator - Blank Template	Fields in blue will be automatically filled in after definition in the next steps.
HTA Submission Number		
Document Version		
Date		
Responsible Team		

Field	What to enter	Use
Title of the Analysis	A concise title identifying the technology and population.	Used to identify the analysis.
HTA Submission Number	Submission, assessment, or internal tracking number, when applicable.	Administrative identification.
Document Version	Version number or revision identifier.	Version control.
Date	Date of the analysis/version.	Version control.
Responsible Team	Team, organization, or analysts responsible for the BIA.	Documentation and accountability.

2. Research question

Describe the target population, the proposed technology or technologies, and the relevant comparators. The information entered here should correspond to the scope of the BIA and should remain consistent with the evidence and cost inputs used later.

2.1. Population

Describe the clinical condition and eligible population. The description should be sufficiently specific to distinguish the modeled population from the broader disease population when eligibility depends on disease severity, biomarker status, treatment line, prior therapy, age, or other restriction.

2. Research question

Describe the target population, interventions, and comparators (alternative treatments or technologies).

2.1. Population

Describe the population under evaluation (clinical condition)

2.2. Proposed technology(ies)

Select the number of technologies under assessment and enter their names. The workbook supports up to four proposed technologies. These are the interventions being evaluated for listing, reimbursement, or introduction into the modeled scenario.

2.2. Proposed Technology(ies)

Describe the technology(ies) under assessment for listing/reimbursement

Number of technologies under assessment

1

Proposed technologies are the "interventions" being evaluated for listing/reimbursement.

Name of the technology

Name of the technology

Name of the technology

Name of the technology

2.2. Proposed Technology(ies)

Describe the technology(ies) under assessment for listing/reimbursement

Number of technologies under assessment

4

Proposed technologies are the "interventions" being evaluated for listing/reimbursement.

Name of the technology

Name of the technology

Name of the technology

Name of the technology

2.3. Comparator(s)

Select the number of comparators and enter their names. The workbook supports up to ten comparators. Comparators represent technologies currently available in the reference scenario. They may include medicines, procedures, tests, or other relevant alternatives.

2.3. Comparator(s)

Describe the technologies currently available for the condition

Number of comparators

1

Comparators are the medications/technologies currently available (on the market) for the condition under analysis.

Name of the comparator

Name of the comparator

Name of the comparator

Name of the comparator

Name of the comparator

Name of the comparator

Name of the comparator

Name of the comparator

Name of the comparator

Name of the comparator

2.4. Assessed scenarios

Assign a name to the reference scenario and to each alternative scenario that will be used. The reference scenario represents the situation without availability of the proposed intervention(s). Up to three alternative scenarios can be defined. These scenarios can represent different

expected uptake patterns, alternative assumptions regarding displacement of comparators, or other scenario-specific market configurations.

2.4. Assessed Scenarios

Provide names to identify the scenarios under evaluation

Name of reference scenario

e.g.: only basic care

The reference scenario refers to the current scenario, in which the interventions are not available.

Name of alternative scenario 1

e.g.: Technology 1 10-50% market uptake

Alternative scenarios refer to the scenarios in which the intervention under analysis will be available.

Name of alternative scenario 2

Name of alternative scenario 3

Different alternative scenarios can be chosen to vary the expected diffusion rates.

Important: At this stage, enter scenario names only. Annual market-share percentages are entered in the Mkt share tab.

3. Market share and time horizon

The Mkt share tab defines the calendar years of the analysis and the annual distribution of eligible patients across technologies in each scenario.

3.1. Time horizon

The current workbook uses a fixed five-year time horizon. Enter the first calendar year of the analysis in YYYY format. The workbook then derives the subsequent four years.

Budget impact analysis of Medic1 for treatment of Disease X - Example

< **Mkt Share & time horizon** > Next: Population

3. Time horizon
Establish the period of the analysis.

Time horizon: **5 years** By definition, this tool evaluates a five-year time horizon.

What is the first year of the analysis? **2027** Write in 20xx format the first year of the budget impact analysis (e.g., 2025).

4. Market Share

Reference scenario

In the reference scenario, the proposed technologies are assumed not to be available and therefore retain a market share of 0% in all five years. Enter the annual market shares of the comparators. The comparator shares should collectively account for 100% of the eligible population in each year.

4. Market share
Establish the diffusion rate of the technology considering the scenarios previously reported (reference scenario and proposed scenarios).
Enter market share assumptions according to the best available evidence and the applicable BIA guidelines.
Linear, non-linear S-shaped, or other uptake patterns may be used, provided they are appropriately justified.

Reference scenario: e.g.: only basic care

	Technology	2027	2028	2029	2030	2031
Intervention	Technology 1					
	Do not fill this line					
	Do not fill this line					
Comparators	Basic care	100,00%	100,00%	100,00%	100,00%	100,00%
	Do not fill this line					
	Do not fill this line					
	Do not fill this line					
	Do not fill this line					
	Do not fill this line					
	Do not fill this line					
	Do not fill this line					
	Do not fill this line					
	Do not fill this line					

100% ok 100% ok 100% ok 100% ok 100% ok

The sum of the percentages must result in 100%

Alternative scenarios

For each alternative scenario, enter the annual market share of each proposed technology and comparator for all five years. The sum across all technologies must equal 100% in every year.

Alternative scenarios may differ in the speed or shape of adoption, in the technologies displaced by the new intervention, or in the final market share reached. The workbook accepts annual user-defined shares and therefore does not require uptake to follow a linear pattern.

Alternative scenario 1: e.g.: Technology 1 10-50% market uptake

Intervention	Technology 1	5,00%	7,00%	10,00%	15,00%	25,00%
	Do not fill this line					
Comparators	Basic care	95,00%	93,00%	90,00%	85,00%	75,00%
	Do not fill this line					
Comparators	Do not fill this line					
	Do not fill this line					
	Do not fill this line					
	Do not fill this line					
	Do not fill this line					
	Do not fill this line					
		100%	100%	100%	100%	100%
		ok	ok	ok	ok	ok

The sum of the percentages must result in 100%

Sources and patterns for market-share assumptions

Market-share assumptions should be based on the best available evidence for the decision context. Potential sources include historical utilization or prescribing data, claims or registry data, observed uptake of analogous technologies, market or administrative data, and structured expert opinion. When empirical evidence is unavailable, assumption-based uptake patterns can be used as scenarios, but they should be clearly identified and justified.

Pattern	Example use	How to enter
Linear uptake	Approximately constant annual increase in adoption.	Enter progressively increasing shares with similar year-to-year increments.
Non-linear uptake	Adoption accelerates or slows over time.	Enter the intended share independently for each year.
S-shaped uptake	Slow initial adoption, faster mid-period diffusion, then stabilization.	Enter low early shares, larger middle-year increments, and smaller later increments.
Immediate/rapid uptake	Technology is expected to gain substantial share soon after introduction.	Enter a relatively high Year 1 share followed by the expected subsequent pattern.
Stable share	No material change in technology mix is expected.	Repeat the same percentage across years.

Important: Market share describes how the eligible population is distributed among technologies; it does not define the size of the eligible population. Population size is estimated separately in the Population tab.

5. Population: estimating the number of eligible patients

The Population tab provides three main pathways: manual data entry, observed demand, and an epidemiological method. Select the pathway that best represents the available evidence. The selected method should produce the total number of patients eligible for allocation across technologies in each budget year.

Initial population questions

Budget Impact Calculator - Blank Template

< **Population** > Next: Costs

5. Estimating the Total Number of Eligible Patients
Select the method to be used to estimate the number of eligible patients.

Calculation method

Enter data manually
Select this option if you want to manually enter the total number of eligible patients, and complete section 5.1. below.

Does the eligible population refer to the number of new cases?

Yes, it represents incidence

Does the reported prevalence refer to a specific line of treatment?

Yes *If Yes, patients who progress in one year will be removed from the eligible prevalent population in the next year.*

Before entering population data, select the calculation method and answer the questions that characterize the population flow.

Input	Meaning	Implication
Calculation method	Manual data entry, observed demand, or epidemiological method.	Determines which population section should be completed.
Does the eligible population refer to the number of new cases?	Identifies whether the annual population represents incidence/newly eligible patients.	Affects whether annual values are interpreted as new cohorts rather than a total prevalent stock.
Does the reported prevalence refer to a specific line of treatment?	Identifies whether progression can remove patients from the eligible prevalent population for the following year.	If Yes, patients who progress in one year can be removed from the eligible prevalent population in the next year.

5.1. Manual data entry

5. Estimating the Total Number of Eligible Patients
Select the method to be used to estimate the number of eligible patients.

Calculation method

Enter data manually
Select this option if you want to manually enter the total number of eligible patients, and complete section 5.1. below.

Enter data manually
Observed demand
Epidemiological method

Use Section 5.1 when the final number of eligible patients has already been estimated outside the workbook or is supplied directly by the decision maker. Enter the eligible population separately for each of the five years. No observed-demand projection or epidemiological funnel is required.

5.1. Manual Data Entry

Use this section if you prefer to manually enter the final estimate of eligible patients, without applying the calculation strategies.

- total number of eligible patients per year	2027	2028	2029	2030	2031


Instructions

This section can be used to report prevalence in year 1 and incidence in subsequent years. In this case, select "Yes, it represents incidence" in the previous question about "Does the eligible population relate to the number of new cases?"

This option is useful when a validated external population model, administrative count, contractual population estimate, or previously calculated epidemiological funnel already provides the final annual eligible population.

Important: When using manual entry, document externally how the annual values were derived. Do not also complete an epidemiological funnel for the purpose of generating the same population, as this can result in double counting or inconsistent population definitions.

Entering a baseline prevalent population followed by incident cases

When the analysis should include the prevalent population in the first year and only new (incident) patients entering in subsequent years, select "Enter data manually" as the population calculation method. Then, select "Yes, it represents incidence" for the question "Does the eligible population refer to the number of new cases?"

5. Estimating the Total Number of Eligible Patients

Select the method to be used to estimate the number of eligible patients.

Calculation method

Enter data manually

Select this option if you want to manually enter the total number of eligible patients, and complete section 5.1. below.

Does the eligible population refer to the number of new cases?

Yes, it represents incidence

In the manual data-entry table, enter the total prevalent eligible population in Year 1 and the number of new eligible patients (incident cases) in each subsequent year. The tool will treat the first-year value as the initial prevalent cohort and subsequent values as new cohorts entering the analysis over time.

For example, if 1,200 patients are eligible at the beginning of the analysis and 20, 19, 20, and 21 new patients become eligible in Years 2–5, respectively, enter:

5.1. Manual Data Entry

Use this section if you prefer to manually enter the final estimate of eligible patients, without applying the calculation strategies.

- total number of eligible patients per year	2027	2028	2029	2030	2031
	1.200	20	19	20	21


Instructions

This section can be used to report prevalence in year 1 and incidence in subsequent years. In this case, select "Yes, it represents incidence" in the previous question about "Does the eligible population relate to the number of new cases?"

The workbook will then account for the initial cohort and the additional incident cohorts according to the analysis settings.

Important: Do not enter the total prevalent population again in each subsequent year when this approach is selected. Doing so would combine a prevalent-stock approach with incident cohort entry and may result in double counting of patients.

5.2. Observed demand

5. Estimating the Total Number of Eligible Patients

Select the method to be used to estimate the number of eligible patients.

Calculation method

Observed demand
Enter data manually
Observed demand
Epidemiological method

Number of eligible patients is estimated based on data obtained from 5 years in the service. Complete section 5.2 below.

Use Section 5.2 when historical service-use or administrative data can be used to project future eligible demand. Examples may include numbers of procedures, treated patients, authorizations, claims, or other utilization records.

- i. Enter the first and last year for which historical data are available.
- ii. Specify the data interval/periodicity requested by the worksheet.
- iii. Describe each historical series. If the estimate depends on more than one procedure or utilization component, enter each component in a separate row.
- iv. Enter the observed values for the available historical years.
- v. Review the historical total and the projection behavior.
- vi. Select the projection strategy: mean projection or linear trend.

[illegible]

The workbook provides both a linear projection and a mean-based projection. The worksheet displays an R²-based prompt indicating that a weak linear fit should prompt consideration of the mean approach. The selected observed-demand estimate is then carried forward to the eligible-population calculation.

An expected annual growth adjustment is also available in the observed-demand section. This can be used when the projected demand should reflect population growth over the analysis period.

Projection calculation strategy: **Mean projection** Select whether the population should be estimated based on the annual average or a linear trend.
 Linear projection: the linear projection is less accurate than the mean projection. You should consider to use the mean.

Population projection for the time horizon of the analysis (linear projection):

	2027	2028	2029	2030	2031
Population projection for the time horizon of the analysis (linear projection)	0	0	0	0	0

Population projection for the time horizon of the analysis (mean projection):

	2027	2028	2029	2030	2031
Population projection for the time horizon of the analysis (mean projection)	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!

*Adjustment for population growth in the period

Expected annual growth: **1,100**

Selected observed demand:

	2027	2028	2029	2030	2031
Mean projection	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!

Important: A statistical fit indicator should not be used as the sole criterion for choosing a projection. The selected approach should also be clinically and operationally plausible and consistent with the data-generating process.

Example: In this example, observed demand data are available for the period from 2010 to 2015, with the number of patients increasing from 200 to 231. The linear projection option is selected to extrapolate the historical trend to the five-year BIA time horizon. An expected annual population growth factor of 1.08 is additionally applied to adjust the projected values for population growth over the period. Based on these inputs, the estimated eligible population is 309 patients in 2027, increasing to 335 patients in 2031. Users should select the projection approach that best reflects the observed historical pattern and the assumptions considered appropriate for the analysis.

5.2. Estimation Using the Observed Demand Method

Use this section if you prefer to estimate the eligible population based on projections from historical data.
 If the estimate depends on more than one procedure, enter the number for each in a separate row.

Instructions

To adjust the charts, click on the chart, click on the top menu "Chart Design" and then click on "Select Data". Make sure that the "Measured Demand" icon is selecting only the fields in the table where there is data (reported period).

Start year of data: **2010** Enter the first year for which data are available.
 End year of data: **2015** Enter the last year for which data are available.
 Data interval: **6**

Description	2010	2011	2012	2013	2014	2015			
Number of patients with the disease	200	203	215	220	225	231			
Sum	200	203	215	220	225	231			

Projection calculation strategy: **Linear projection** Select whether the population should be estimated based on the annual average or a linear trend.
 Tip: If the R^2 value of the linear projection is less than 0.5, you should consider to use the mean.

Population projection for the time horizon of the analysis (linear projection)	2027	2028	2029	2030	2031
	309	316	322	329	335

Population projection for the time horizon of the analysis (mean projection)	2027	2028	2029	2030	2031
	216	216	216	216	216
	233	252	272	293	317

*Adjustment for population growth in the period

Selected observed demand	2027	2028	2029	2030	2031
Linear projection	309	316	322	329	335

Expected annual growth: **1,080**

5.3. Epidemiological method

5. Estimating the Total Number of Eligible Patients

Select the method to be used to estimate the number of eligible patients.

Calculation method

Epidemiological method

Enter data manually: population data are used to estimate the total number of eligible patients (e.g. prevalence). Complete section 5.3 below.

Observed demand

Epidemiological method

Use Section 5.3 when the eligible population is estimated by applying epidemiological or eligibility parameters to a base population.

5.3. Estimation Using the Epidemiological Method

Use this section if you wish to estimate the eligible population based on the epidemiological parameters from the literature.

This section can also be used if you wish to apply epidemiological parameters to the population projected through the "Observed Demand" method.

Parameter	Value (%)	2027	2028	2029	2030	2031	Total
	-	0	0	0	0	0	0
		0	0	0	0	0	0
		0	0	0	0	0	0
		0	0	0	0	0	0
		0	0	0	0	0	0
		0	0	0	0	0	0
		0	0	0	0	0	0
		0	0	0	0	0	0
Final		0	0	0	0	0	0

The epidemiological section can also be applied after the observed-demand method when the observed-demand projection represents an upstream population that still requires additional eligibility filters:

5.3. Estimation Using the Epidemiological Method

Use this section if you wish to estimate the eligible population based on the epidemiological parameters from the literature.

This section can also be used if you wish to apply epidemiological parameters to the population projected through the "Observed Demand" method.

Parameter	Value (%)	2027	2028	2029	2030	2031	Total	Reference
Observed demand - disease	-	309	316	322	329	335	1,611	Previous example of observed demand
Percentage of serious disease	40,000%	124	126	129	131	134	644	
		0	0	0	0	0	0	
		0	0	0	0	0	0	
		0	0	0	0	0	0	
		0	0	0	0	0	0	
		0	0	0	0	0	0	
		0	0	0	0	0	0	
		0	0	0	0	0	0	
Final		124	126	129	131	134		

For each epidemiological parameter, enter the parameter description, percentage/value, and reference where applicable. The workbook sequentially applies the entered parameters to derive the final eligible population.

Potential parameters depend on the decision problem and may include disease prevalence or incidence, age or sex restrictions, disease severity, biomarker status, diagnosis rate, treatment eligibility, treatment-line eligibility, contraindications, or other clinically relevant filters.

Incidence, prevalence, and prevalent-stock handling

The distinction between incident and prevalent populations is essential because the annual values represent different cohort structures.

Population approach	Interpretation	Recommended use
Annual incident/newly eligible cases	Each year's input represents new patients entering eligibility during that year.	Use when the decision problem concerns newly diagnosed/newly eligible patients or when incidence is the natural flow measure.
Baseline prevalent stock + incident flow	Year 1 includes the existing eligible stock; later years add newly eligible	Use when existing prevalent patients remain eligible over

	incident cases while accounting for relevant attrition from the prior stock.	time and new cases enter the population.
Repeated annual prevalence estimate	Each year's prevalence estimate already represents the total eligible stock for that year.	Use when a reliable annual prevalence estimate is directly available and already incorporates the population stock for each year.

Important: Do not combine a baseline prevalent stock plus incident flow with a repeated annual prevalence estimate for the same population. These are alternative ways of representing the population stock. Combining them can count continuing prevalent patients twice. If annual prevalence values already represent the total eligible stock in each year, do not add incident cases again unless the prevalence estimate explicitly excludes them.

When prevalence refers to a specific treatment line and progression removes patients from that line, select the corresponding option in the workbook so that progression in one year can reduce the eligible prevalent population carried into the next year.

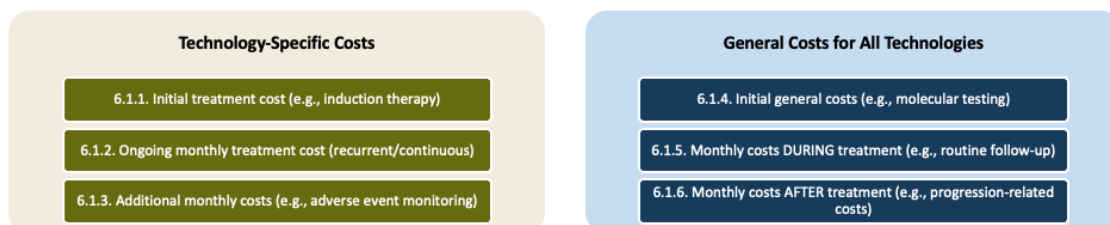
Missing or incomplete epidemiological evidence

If no single source provides all required epidemiological parameters, use the best available evidence for each parameter, ensuring that definitions and denominators are compatible. Clearly document extrapolations, assumptions, or substitutions. Parameters with substantial uncertainty should be explored through deterministic or probabilistic sensitivity analysis or through alternative scenarios when appropriate.

6. Costs

The Costs tab estimates per-patient costs using a structured set of technology-specific and general cost categories. The summary table at the top of the tab is automatically populated from the detailed sections and should not be edited directly.

Cost structure



Cost category	Scope	Timing
Initial treatment cost	Technology-specific	One-time cost when the patient enters treatment.
Ongoing monthly treatment cost	Technology-specific	Recurrent/continuous treatment cost.
Additional ongoing monthly costs	Technology-specific	Other recurrent costs associated with a specific technology.
Initial general costs	Common to all technologies	One-time cost at entry, e.g., diagnostic or molecular testing.
Monthly costs during treatment	Common to all technologies	Recurrent costs while the patient remains on treatment.
Monthly costs after treatment	Common to all technologies	Recurrent costs after treatment discontinuation, e.g., costs related to progression.

Resource-use fields

Field	How to use it
Select the technology	Choose the intervention/comparator to which the resource applies. General-cost sections are applied across technologies.
Description of resource	Describe the medicine, procedure, test, visit, administration, monitoring item, or other resource.
% of patients using	Enter the proportion of patients expected to use the resource.
Quantity	Enter the number of units required for the specified costing period.
Unit cost	Enter the cost per unit in the relevant currency and price basis.
Final cost	Automatically calculated from the entered resource-use components.
Reference	Document the source of the resource-use or unit-cost estimate.

6.1.1. Initial treatment costs

6.1.3. Associated monthly treatment-related costs

Use Section 6.1.3 for **recurrent costs that are specifically associated with a given intervention or comparator** and accrue while the patient is receiving that technology. These costs should be distinguished from the acquisition or administration cost of the technology itself and from routine costs that are common to all treatment alternatives.

Examples may include technology-specific laboratory monitoring, periodic imaging or other examinations required because of a particular treatment, specialist visits required only for patients receiving that technology, premedication administered repeatedly with treatment, or recurrent management of treatment-related adverse events when these costs are expected to occur throughout treatment.

For each cost item, users must select the technology to which the resource use applies. Multiple rows may be entered for the same technology, allowing different resources, frequencies, and unit costs to be represented separately. For example, a treatment requiring both monthly laboratory monitoring and a specialist consultation every three months may include these as separate cost items assigned to the same technology.

6.1.4. Initial general costs

Use Section 6.1.4 for **one-time costs incurred at the beginning of the modeled pathway that apply to all eligible patients**, regardless of the technology received. Because these costs are not treatment-specific, they should not be assigned separately to individual interventions or comparators.

Examples include an initial diagnostic assessment required for all eligible patients, a molecular or genetic test used to confirm eligibility, baseline imaging, an initial specialist consultation, or other examinations performed once before treatment selection.

Costs that apply only to patients receiving a particular technology should instead be entered as technology-specific costs, even if they occur only once.

6.1.5. Monthly costs during treatment

Use Section 6.1.5 for **recurrent healthcare costs that are common to patients across all modeled technologies** and accrue while patients remain on treatment. This category is appropriate when the type and frequency of resource use are assumed not to differ according to the intervention or comparator received.

Examples include routine clinical follow-up, standard laboratory monitoring performed for all treated patients, periodic general medical consultations, supportive care, or other recurring healthcare resources required as part of the disease-management pathway independently of the specific technology used.

If the frequency or cost of a resource differs by technology, it should generally be entered under Associated monthly treatment-related costs (Section 6.1.3) rather than as a general monthly cost.

6.1.6. Monthly costs after treatment

Use Section 6.1.6 for **recurrent healthcare costs that begin after a patient discontinues treatment** and are expected to accrue during the post-treatment period. The workbook treats

these costs separately from costs incurred while patients are receiving treatment, allowing changes in resource use following treatment discontinuation to be represented in the BIA.

Examples may include routine follow-up after treatment cessation, disease-management costs incurred after discontinuation, monitoring of patients no longer receiving active treatment, supportive care, or other recurrent healthcare resources associated with the post-treatment period.

These costs should be included only when a post-treatment resource-use pattern is relevant to the analysis. Costs that continue only while the patient remains on treatment should instead be entered under the appropriate treatment-period cost category.

6.2. Cost Support Table

Section 6.2 is an alternative input pathway for situations in which the structured cost sections cannot reasonably represent the available cost data. Select the Cost Support Table option only when required. If the structured Section 6.1 tables are used, keep this option set to No.

The support table accepts final technology-level cost estimates for the same broad cost categories: initial treatment cost, ongoing annual treatment cost, additional ongoing annual costs, initial general costs, annual costs during treatment, and annual costs after treatment.

6.2. Cost Support Table

Use this section only if it is not possible to estimate the costs of each intervention using the structured steps above. These values should be used only as a last resort, when calculating costs through the standard tables is not feasible.

Using the 'Cost Support Table' option, please keep "No" selected if costs are being calculated using the Section 6.1 tables.

Technology	Technology assessed	Technology-specific Costs			General Costs (common to all patients)		
		Initial treatment cost (one-time)	Ongoing annual cost	Additional ongoing annual costs	Initial general costs	Annual costs during treatment	Annual costs after treatment
Intervention 1	Technology assessed						
Intervention 2	Do not fill this line				\$ -	\$ -	\$ -
Intervention 3	Do not fill this line				\$ -	\$ -	\$ -
Intervention 4	Do not fill this line				\$ -	\$ -	\$ -
Comparator 1	Comparator				\$ -	\$ -	\$ -
Comparator 2	Do not fill this line				\$ -	\$ -	\$ -
Comparator 3	Do not fill this line				\$ -	\$ -	\$ -
Comparator 4	Do not fill this line				\$ -	\$ -	\$ -
Comparator 5	Do not fill this line				\$ -	\$ -	\$ -
Comparator 6	Do not fill this line				\$ -	\$ -	\$ -
Comparator 7	Do not fill this line				\$ -	\$ -	\$ -
Comparator 8	Do not fill this line				\$ -	\$ -	\$ -
Comparator 9	Do not fill this line				\$ -	\$ -	\$ -
Comparator 10	Do not fill this line				\$ -	\$ -	\$ -

Important: The Cost Support Table should be used as a fallback rather than in parallel with the structured cost tables. Using both pathways for the same cost components may double count costs.

7. Time on treatment and patient attrition

The Time on treatment tab controls how long costs accrue for each technology and, where relevant, how patients discontinue treatment or exit the analysis because of death. It separates the planned duration of treatment from discontinuation/progression and mortality.

7.1. Label-based treatment duration

In Section 7.1, enter the number of months in each of the five years for which each intervention and comparator is expected to be used according to the treatment regimen or product label.

This input describes the planned treatment schedule only. It does not itself account for early discontinuation due to progression, adverse events, lack of efficacy, or other reasons; those mechanisms are entered separately.

Budget Impact Calculator - Blank Template

< Time on treatment > Results

7.1 Reporting Treatment Duration According to the Product Label

Use this section to specify the number of months per year that each technology is expected to be used, based on the approved product label.

This duration refers solely to the recommended usage and does not account for discontinuation due to disease progression or other reasons (which will be reported in the following section). If the treatment includes both an induction and a maintenance phase, report only the number of months recommended for the maintenance phase.

For how many months will each intervention be used per patient?

Informe a resposta para cada ano

	1st year	2nd year	3rd year	4th year	5th year	
Do not fill this line:						months
Do not fill this line:						months
Do not fill this line:						months
Do not fill this line:						months

For how many months will each comparator be used per patient?

Informe a resposta para cada ano

	1st year	2nd year	3rd year	4th year	5th year	
Do not fill this line:						months
Do not fill this line:						months
Do not fill this line:						months
Do not fill this line:						months
Do not fill this line:						months
Do not fill this line:						months
Do not fill this line:						months
Do not fill this line:						months
Do not fill this line:						months
Do not fill this line:						months

Important: If treatment includes induction and maintenance phases, the workbook instructs users to report only the number of months recommended for the maintenance phase in this section because induction costs can be entered separately as initial treatment costs.

The annual month inputs allow different patterns, including treatment restricted to Year 1, treatment continuing for a fixed number of years, or treatment continuing throughout the five-year horizon.

Example: In this example, Technology 1 is assumed to be a continuous treatment and is therefore used for 12 months in each year of the five-year time horizon. Technology 2 has a fixed treatment duration of two years, so 12 months of treatment are entered in Years 1 and 2 and zero months thereafter. The comparator represents a six-cycle chemotherapy regimen completed within six months, and therefore 6 months are entered in Year 1 and zero months in subsequent years. This section allows treatment duration to differ across technologies and over time, ensuring that recurrent treatment-related costs are applied only during the period in which each technology is expected to be used.

For how many months will each intervention be used per patient?

	1st year	2nd year	3rd year	4th year	5th year	
Technology 1 (continuous)	12	12	12	12	12	months
Technology 2 (2 years)	12	12	0	0	0	months
Do not fill this line						months
Do not fill this line						months

For how many months will each comparator be used per patient?

	1st year	2nd year	3rd year	4th year	5th year	
Chemo - 6 cycles	6	0	0	0	0	months
Do not fill this line						months
Do not fill this line						months
Do not fill this line						months
Do not fill this line						months
Do not fill this line						months
Do not fill this line						months
Do not fill this line						months
Do not fill this line						months
Do not fill this line						months

7.2. Treatment discontinuation

Section 7.2 is optional. Select No when patients are expected to remain on treatment for the full duration specified in Section 7.1, subject only to any mortality adjustment selected later. Select Yes when treatment discontinuation or progression should reduce the number of patients remaining on treatment.

7.2. Estimating Patient Discontinuation and Number of Patients on Treatment

Use this section if it is important to indicate the percentage of patients who remain on treatment or discontinue treatment over the course of the year. Treatment discontinuation can be estimated using different parameters: median treatment duration, discontinuation rate, or disease progression.

Do you wish to consider treatment discontinuation?

Yes ☒ No ☐

If all patients are expected to remain on treatment for the entire period specified in the previous section, or until death (next section).



Information

Median treatment duration: This refers to the time point at which 50% of patients have either completed or discontinued treatment (expressed in months).

Treatment discontinuation rates: These represent the proportion of patients who stop treatment in a given year, whether due to adverse events, lack of efficacy, or other reasons.

Progression-free survival (PFS): This is the percentage of patients who remain free from disease progression after initiating treatment. It should be used only when progression is a proxy or explicit treatment-stopping criterion.

The workbook allows treatment discontinuation to be represented using different evidence types, including median treatment duration, discontinuation rates, or disease progression/progression-free survival.

Input type	Interpretation	Use
Median usage time	Time at which approximately 50% of patients have discontinued/completed treatment under the modeled assumption.	Enter median time in months for each relevant technology.
Discontinuation rate	Proportion/cumulative proportion discontinuing treatment according to the selected data structure.	Enter the available discontinuation data at the selected periodicity.

Disease progression / PFS	Progression information used as a proxy for treatment stopping.	Use only when progression is a relevant criterion for treatment discontinuation.
---------------------------	---	--

The worksheet also asks whether discontinuation data include deaths. This selection is important to avoid representing the same mortality-related attrition twice when mortality is also modeled separately.

Data periodicity for discontinuation/progression

Select the data period that corresponds to the available evidence. Annual data can be entered in the main Time on treatment tab. When another periodicity is selected, the Data tab provides supporting tables for more granular discontinuation or progression-free-survival information, including monthly or semi-annual data structures.

The Data tab also contains the workbook's internal extrapolation of median treatment duration using an exponential distribution. Users should not overwrite these calculation cells.

7.3. Exit due to death

Section 7.3 is optional and should be used when mortality materially affects the number of patients remaining in the analysis. Select No when mortality is not modeled separately. Select Yes when patients should exit the modeled population because of death.

The workbook supports mortality information through median survival or survival/mortality data. The current worksheet includes options for cumulative mortality rate and overall-survival-type inputs and allows selection of the data period.

Input	Interpretation
Median survival	Time in months at which 50% of the relevant population is alive; the workbook uses the corresponding internal extrapolation structure.
Cumulative mortality rate	Cumulative proportion of patients who have died by each time point; values should be non-decreasing over time.
Overall survival	Proportion of patients alive at each time point; values should be non-increasing over time.
Other data periodicity	Use the supporting Data tab when the evidence is available at a frequency other than the annual structure shown in the main tab.

Important: Ensure consistency between discontinuation and mortality inputs. If the discontinuation series already includes deaths and mortality is also modeled separately, review the selected workbook option carefully to avoid double counting patient exits.

8. Results

The Results tab summarizes the population and budget consequences of each alternative scenario relative to the reference scenario. The workbook presents results for each of the five budget years and summary measures such as cumulative and annual mean values.

8.1. Final population

Review the final population used in the calculations before interpreting cost results. This provides an important check that the selected population method, market shares, and attrition assumptions have produced a plausible number of patients.

The Results tab includes an option asking whether dynamic cohorts are required for population projection. When dynamic cohorts are used, patients can transition between cohorts over time based on the treatment-duration/attrition information provided in the workbook. This option should be selected only when the cohort structure of the analysis requires patients entering in different years to be followed across subsequent years.

Budget Impact Calculator - Blank Template

< Results > Next: Determinist Sensibility Analysis

8. Results

	2027	2028	2029	2030	2031
8.1. Final population used in the calculations	0	0	0	0	0

Is it necessary to use dynamic cohorts to project the population? ☐ No ☒ Yes

Yes, patients will transition between cohorts based on the data to be provided in the "time to treatment" tab.

8.2. Full-cost results

Full-cost results incorporate the technology-specific and general cost components represented in the model. For each alternative scenario, the workbook reports the total budget for the reference and alternative scenarios and the incremental budget impact.

8.3. Technology cost-only results

The workbook also reports a technology cost-only perspective, isolating technology-related acquisition/treatment costs from the broader set of modeled associated costs. This output can be useful as a complementary scenario, but it should not replace the full-cost result when the decision problem requires all relevant budget consequences.

Interpreting incremental budget impact

Incremental budget impact is the difference between the alternative scenario and the reference scenario for the corresponding budget year or cumulative period. A positive value indicates higher modeled expenditure under the alternative scenario, whereas a negative value indicates lower modeled expenditure relative to the reference scenario.

Alternative scenario 1: Drug 1 only

Adoption rate ranging from 5% to 25%

Full-cost results							Annual mean	
	2026	2027	2028	2029	2030	Cumulative		
Reference scenario	\$ 8.174.700	\$ 15.386.109	\$ 22.651.801	\$ 29.972.284	\$ 37.348.729		\$	22.706.725
Alternative scenario 1	\$ 9.333.685	\$ 16.545.125	\$ 23.811.786	\$ 31.293.218	\$ 38.669.158		\$	27.795.051
Incremental budget impact	\$ 1.158.185	\$ 3.417.067	\$ 5.385.539	\$ 7.056.463	\$ 8.424.378	\$ 25.441.632	\$	5.088.326
Mean	\$ 1.155.333	\$ 3.408.185	\$ 5.376.334	\$ 7.048.305	\$ 8.423.583	\$ 25.400.816		
95% CI (lower)	\$ 653.917	\$ 1.926.287	\$ 2.977.089	\$ 3.793.733	\$ 4.362.713	\$ 13.731.222		
95% CI (upper)	\$ 1.656.525	\$ 4.890.684	\$ 7.773.372	\$ 10.303.058	\$ 12.482.462	\$ 37.088.617		

Technology cost-only results							Annual mean	
	2026	2027	2028	2029	2030	Cumulative		
Reference scenario	\$ 6.427.969	\$ 12.906.419	\$ 19.433.897	\$ 26.010.622	\$ 32.637.540		\$	19.483.289
Alternative scenario 1	\$ 7.586.954	\$ 14.067.899	\$ 20.594.874	\$ 27.171.597	\$ 33.798.515		\$	24.583.689
Incremental budget impact	\$ 1.158.185	\$ 3.418.768	\$ 5.392.387	\$ 7.073.679	\$ 8.458.980	\$ 25.501.998	\$	5.100.400
Mean	\$ 1.155.333	\$ 3.410.886	\$ 5.384.539	\$ 7.065.510	\$ 8.458.185	\$ 25.460.193		
95% CI (lower)	\$ 663.473	\$ 1.955.416	\$ 3.019.641	\$ 3.844.051	\$ 4.415.673	\$ 13.917.112		
95% CI (upper)	\$ 1.660.931	\$ 4.905.885	\$ 7.803.725	\$ 10.355.800	\$ 12.567.747	\$ 37.275.231		

Budget-impact outputs are conditional on the population, uptake, treatment-duration, cost, and attrition assumptions entered by the user. They should be interpreted as scenario-based estimates rather than as guaranteed future expenditure.

Deterministic sensitivity analysis (DSA)

The workbook includes a deterministic sensitivity analysis module. DSA evaluates the effect of changing selected uncertain parameters individually while holding the remaining parameters at their base-case values.

Uncertainty settings are managed through the Parameters structure. Parameters can be assigned a base-case estimate, a variation or explicit lower/upper bounds, a reference, and an indicator specifying whether the parameter is included in the DSA.

Defining uncertainty ranges

The current parameter structure includes fields for Variation, SE, Min, and Max. A percentage variation can be used to generate lower and upper bounds around the base-case value. Alternatively, lower and upper bounds from source data can be entered when available.

When lower and upper bounds represent an approximately symmetric 95% confidence interval, the workbook derives the standard error as (upper bound – lower bound) / 3.92. When the bounds are assumption-based rather than empirical confidence limits, they should be described as assumption-based uncertainty bounds in the analysis documentation.

Parameters	Code	In use	Base line	Probability	Variation	SE	Min	Max	Distribution
Incidence of cancer		0,00	0,000031	#DIV/0!	20%	0,0000			Beta
Advanced disease		0,25	0,250000	0,0000	20%	0,0000			Beta
X mutation		0,50	0,500000	0,4624	20%	0,0510			Beta
0		0,00	0,000000	#DIV/0!	20%	0,0000			Beta
0		0,00	0,000000	#DIV/0!	20%	0,0000			Beta
0		0,00	0,000000	#DIV/0!	20%	0,0000			Beta
0		0,00	0,000000	#DIV/0!	20%	0,0000			Beta
0		0,00	0,000000	#DIV/0!	20%	0,0000			Beta
Initial costs Drug 1	ci_int1	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma
Initial costs Drug 2	ci_int2	200,00	\$200,00	195,7586	20%	20,4082			Gamma
Initial costs	ci_int3	0,00	\$0,00	0,0000	20%	0,0000			Gamma
Initial costs	ci_int4	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma
Initial costs BSC	ci_comp1	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma
Initial costs	ci_comp2	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma
Initial costs	ci_comp3	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma
Initial costs	ci_comp4	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma
Initial costs	ci_comp5	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma
Initial costs	ci_comp6	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma
Initial costs	ci_comp7	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma
Initial costs	ci_comp8	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma
Initial costs	ci_comp9	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma

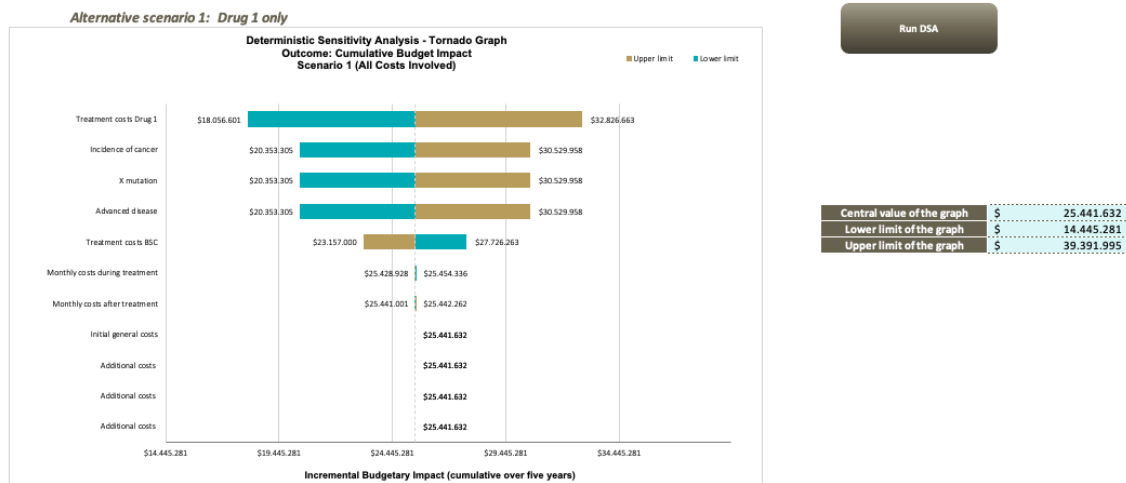
DSA outputs

The SensibilityAn tab displays DSA results and graphs for the modeled scenario outputs. The DSA support calculations include lower and upper parameter limits for full-cost and technology cost-only results.

Budget impact analysis of treatments for Advanced cancer

< Sensitivity analysis > Next: Probabilistic Sensitivity Analysis

9. Deterministic sensitivity analysis (DSA)



Probabilistic sensitivity analysis (PSA)

The PSA module simultaneously varies uncertain parameters according to probability distributions and propagates parameter uncertainty through the budget impact calculations. The current workbook is structured for 1,000 simulation iterations.

Parameters and distributions

The Parameters tab identifies parameters used in the uncertainty analyses. Epidemiological probabilities or proportions are assigned beta distributions by default, while cost parameters are assigned gamma distributions. A normal distribution can also be selected when appropriate for the parameter being modeled.

Distribution	Typical use in this workbook	Parameterization
Beta	Probabilities/proportions bounded between 0 and 1.	Derived from the point estimate and standard error.
Gamma	Positive cost parameters.	Derived from the mean and standard error.
Normal	Parameters for which an approximately symmetric unbounded distribution is considered appropriate.	Mean = point estimate; dispersion based on the standard error.

- For a beta distribution with mean p and standard error s , the workbook parameterization corresponds to $\alpha = p \times [p(1-p)/s^2 - 1]$ and $\beta = (1-p) \times [p(1-p)/s^2 - 1]$.
- For a gamma distribution with mean μ and standard error s , the corresponding shape is μ^2/s^2 and scale is s^2/μ .

Important: Distribution choice should reflect the mathematical support and plausible range of the parameter. In particular, a normal distribution can generate values outside natural bounds and should therefore be selected only when appropriate.

Budget impact analysis of treatments for Advanced cancer

Parameters	Code	In use	Base line	Probabilistic	Variation	SE	Min	Max	Distribution	Alpha	Beta
Incidence of cancer		0,00	0,000031	#DIV/0!	20%	0,0000			Beta	#DIV/0!	#DIV/0!
Advanced disease		0,25	0,250000	0,3240	20%	0,0255			Beta	96,04	3097871,44
X mutation		0,50	0,500000	0,4624	20%	0,0510			Beta	71,78	215,34
0		0,00	0,000000	#DIV/0!	20%	0,0000			Gamma	1	47,52
0		0,00	0,000000	#DIV/0!	20%	0,0000			Beta	#DIV/0!	#DIV/0!
0		0,00	0,000000	#DIV/0!	20%	0,0000			Beta	#DIV/0!	#DIV/0!
0		0,00	0,000000	#DIV/0!	20%	0,0000			Beta	#DIV/0!	#DIV/0!
0		0,00	0,000000	#DIV/0!	20%	0,0000			Beta	#DIV/0!	#DIV/0!
Initial costs Drug 1	ci_int1	0,00	\$0,00	#DIV/0!	20%	0,0000			Beta	#DIV/0!	#DIV/0!
Initial costs Drug 2	ci_int2	200,00	\$200,00	195,7586	20%	20,4082			Gamma	96,04	2,08
Initial costs	ci_int3	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma	#DIV/0!	#DIV/0!
Initial costs	ci_int4	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma	#DIV/0!	#DIV/0!
Initial costs BSC	ci_comp1	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma	#DIV/0!	#DIV/0!
Initial costs	ci_comp2	0,00	\$0,00	#DIV/0!	20%	0,0000			Gamma	#DIV/0!	#DIV/0!

Random sampling

Parameters are sampled independently in the current implementation; correlations between parameters are not modeled. Random draws are generated using the spreadsheet's built-in random-number generator, and the current implementation does not use a fixed random seed.

PSA outputs

The PSA tab stores simulation results for the reference scenario, alternative scenario, and scenario difference. Results are shown by budget year and cumulatively. The tab summarizes the mean simulated budget impact and an uncertainty interval derived from the simulated distribution for each year and the cumulative five-year period.

Because no fixed seed is used, repeated PSA runs may produce slightly different simulation summaries. This is expected for Monte Carlo simulation.

Alternative scenario 1: Drug 1 only								
Adoption rate ranging from		5%	to		25%			
Full-cost results		2026	2027	2028	2029	2030	Cumulative	Annual mean
Reference scenario:		\$ 8.174.700	\$ 15.386.109	\$ 22.651.801	\$ 29.972.284	\$ 37.348.729		\$ 22.706.725
Alternative scenario 1:		\$ 9.332.885	\$ 18.803.176	\$ 28.037.340	\$ 37.028.747	\$ 45.773.106		\$ 27.795.051
Incremental Mean		\$ 1.158.185	\$ 3.416.967	\$ 5.385.539	\$ 7.046.463	\$ 8.424.377	\$ 25.409.919	\$ 5.088.326
95% CI (lower)		\$ 653.917	\$ 1.926.287	\$ 2.977.089	\$ 3.793.733	\$ 4.362.713	\$ 13.731.222	
95% CI (upper)		\$ 1.656.525	\$ 4.890.684	\$ 7.773.372	\$ 10.303.058	\$ 12.482.462	\$ 37.088.617	
Technology cost-only results		2026	2027	2028	2029	2030	Cumulative	Annual mean
Reference scenario:		\$ 6.427.969	\$ 12.906.419	\$ 19.433.897	\$ 26.010.622	\$ 32.637.540		\$ 19.483.289
Alternative scenario 1:		\$ 7.586.155	\$ 16.325.186	\$ 24.826.284	\$ 33.084.300	\$ 41.096.520		\$ 24.583.689
Incremental Mean		\$ 1.158.185	\$ 3.416.967	\$ 5.385.539	\$ 7.046.463	\$ 8.424.377	\$ 25.409.919	\$ 5.100.400
95% CI (lower)		\$ 663.473	\$ 1.955.416	\$ 3.019.641	\$ 3.844.051	\$ 4.415.673	\$ 13.917.112	
95% CI (upper)		\$ 1.660.931	\$ 4.905.885	\$ 7.803.725	\$ 10.355.800	\$ 12.567.747	\$ 37.275.231	

Supporting and calculation tabs

Several tabs support calculations but are not part of the routine user-entry sequence.

Tab	Function
Data	Supports monthly/semi-annual discontinuation and progression-free-survival inputs and internal exponential extrapolation from median treatment/survival information.
Calculations	Tracks cohort entry and within-horizon calculations across technologies and years.
Calculations2	Applies market share, treatment persistence, and cost calculations, including full-cost and technology cost-only structures.
Calculations3	Additional scenario/technology aggregation calculations.
Parameters	Centralizes base-case values and uncertainty parameterization used in DSA and PSA.
DSA	Stores lower/upper-limit calculations used to generate deterministic sensitivity-analysis results.

Important: Do not modify calculation cells in these tabs during routine use. If the model architecture is intentionally modified, the modified workbook should undergo renewed computational checks before use.

Recommended quality-control checks before reporting results

- Confirm that the target population, proposed technologies, comparators, and scenarios match the stated decision problem.
- Confirm that the first analysis year is correct and that the five-year horizon is appropriate for the intended use.
- Check that market shares sum to 100% in every year and scenario and that proposed technologies remain at 0% in the reference scenario.
- Verify that the selected population method matches the available evidence and that incident and prevalent population concepts have not been combined inconsistently.
- For prevalence-based analyses, confirm whether annual prevalence represents a repeated total stock or whether a baseline prevalent stock plus incident flow is being modeled.
- Check all epidemiological percentages for denominator consistency and document the source of each parameter.
- Review resource quantities, utilization proportions, unit costs, price year, and costing perspective.
- Confirm that the Cost Support Table is not being used in parallel with structured cost inputs for the same cost components.
- Confirm that treatment duration reflects the regimen and that induction costs are not duplicated between initial and recurring cost sections.
- Check whether discontinuation data include death and ensure mortality is not double counted.
- Review the final modeled population before reviewing budget results.
- Review both annual and cumulative incremental budget impact and verify the direction of the difference.
- Identify assumption-based market share, epidemiological, cost, or uncertainty bounds explicitly.
- Run DSA and/or PSA for parameters that materially influence the result or are substantially uncertain.
- Save the completed workbook as a separate version and retain a record of the sources and assumptions used.

Suggested reporting of key workbook inputs

For reproducibility, a BIA report prepared with this workbook should describe at minimum the items below. The workbook itself should not be treated as a substitute for transparent reporting.

Domain	Minimum information to report
Decision problem	Population, intervention(s), comparator(s), perspective/context, and scenarios.
Time horizon	First year and five-year period.

Population	Estimation method, incidence/prevalence interpretation, parameters, sources, and cohort-flow assumptions.
Market share	Annual shares by scenario, source/rationale, and whether uptake is empirical, expert-informed, or assumption-based.
Costs	Cost categories, resource use, unit costs, price basis/year, and sources.
Treatment duration	Label/regimen assumptions, discontinuation/progression method, data periodicity, and sources.
Mortality	Whether modeled; source and type of survival/mortality input.
Uncertainty	Parameters varied, bounds, distributions, and scenario assumptions.
Results	Annual and cumulative costs and incremental budget impact; specify full-cost versus technology cost-only results.

Troubleshooting and common modeling errors

Issue	Likely cause / check
Market-share total differs from 100%	One or more technology shares are missing or the displacement of comparators has not been fully allocated.
Eligible population appears too high	Check whether prevalence and incidence were combined twice, whether sequential epidemiological filters use compatible denominators, and whether annual prevalence already represents the total stock.
Costs appear duplicated	Check whether the same resource was entered as both initial and recurring, or in both structured cost tables and the Cost Support Table.
Treatment costs continue too long	Review months of treatment by year and discontinuation/progression settings.
Patient counts fall unexpectedly	Review discontinuation, progression, mortality, and whether discontinuation data already include deaths.
PSA produces implausible draws	Review distribution choice, bounds, standard error, and whether the parameter is compatible with the selected distribution.
Results change slightly between PSA runs	The workbook uses random sampling without a fixed seed; Monte Carlo summaries can vary between runs.
Output does not match an external BIA	First harmonize population definitions, time horizon, market shares, cost components, treatment-duration assumptions, and rounding before interpreting the discrepancy as a computational difference.

Appendix A. Input map by workbook tab

Tab	Principal user inputs
Definitions	Analysis title; submission number; version; date; responsible team; target population; number/names of proposed technologies; number/names of comparators; scenario names.
Mkt share	First analysis year; annual market shares for comparators in the reference scenario and for all technologies in up to three alternative scenarios.

Population	Calculation method; incidence/prevalence characterization; manual annual population OR observed-demand data/projection OR epidemiological parameters and references.
Costs	Technology selection; resource descriptions; utilization proportions; quantities; unit costs; references; optional Cost Support Table.
Time on treatment	Months of treatment per year; whether discontinuation is modeled; discontinuation method; data periodicity; median treatment time/discontinuation/PFS inputs; whether mortality is modeled; mortality method and periodicity.
Results	Dynamic-cohort selection where applicable; otherwise primarily output review.
Parameters	Variation; lower/upper bounds; distribution; reference; DSA inclusion; other uncertainty settings.
Data	Granular discontinuation/PFS/mortality data when the selected periodicity requires the support table.

Appendix B. Glossary

Term	Definition in the context of this workbook
Budget impact	Difference in modeled expenditure between an alternative scenario and the reference scenario over the specified budget period.
Reference scenario	Scenario representing current practice without availability of the proposed technology or technologies.
Alternative scenario	Scenario in which one or more proposed technologies become available and market shares are redistributed.
Market share	Proportion of the eligible population allocated to each technology in a given year and scenario.
Incident population	New cases/newly eligible patients entering during a defined period.
Prevalent population	Existing stock of eligible patients at a defined point/period.
Observed demand	Population estimate based on historical utilization or service data.
Epidemiological method	Population estimate based on a base population and sequential epidemiological/eligibility parameters.
Dynamic cohort	Cohort structure in which patients entering in different periods can remain in or transition through subsequent periods.
Full-cost result	Budget result including the modeled technology-specific and general cost components.
Technology cost-only result	Budget result focusing on technology-related treatment/acquisition costs.
DSA	Deterministic sensitivity analysis: one-at-a-time variation of selected parameters.
PSA	Probabilistic sensitivity analysis: simultaneous random sampling of uncertain parameters from specified distributions.